Binary classification (yes or no) and how to measure it

Can a binary classifier say yes or no to "overheating", and how do we count its true and false positives and negatives?

CML3111 | Electrical, Mechatronics, Robotics and Communications | Week 9, class 2 of 4 | Friday 13 November 2026, 11:00 to 14:00
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Today in one view: what you will leave with

By the end of today you can



Turn the motor winding temperature into a binary label (positive class or negative class) with a limit



Train logistic regression and a decision tree classifier, and work out a confusion matrix and four scores by hand



Move the decision threshold and explain precision, recall, F1, the ROC curve and AUC in plain words, with the formulas

Your three hours: ideas first, then the lab, then the quiz

0:00	10	Recall from Class 1
0:10	35	Concepts 1: labels, z-scores, sigmoid
0:45	30	Concepts 2: confusion matrix, thresholds
1:15	15	Break
1:30	70	Lab: binary classification, in the notebook
2:40	15	Quiz opens: five questions, open 24 hours
2:55	5	Close and what is assessed

Concepts first, with no laptops. Then the lab, then a short quiz.

Three questions from last time

1. What is a baseline model for?

2. What does the unit of the MAE tell you?

3. Why do we split the rows by equipment group?



Answer aloud first. Then we check together.

Answers to the three questions from Class 1

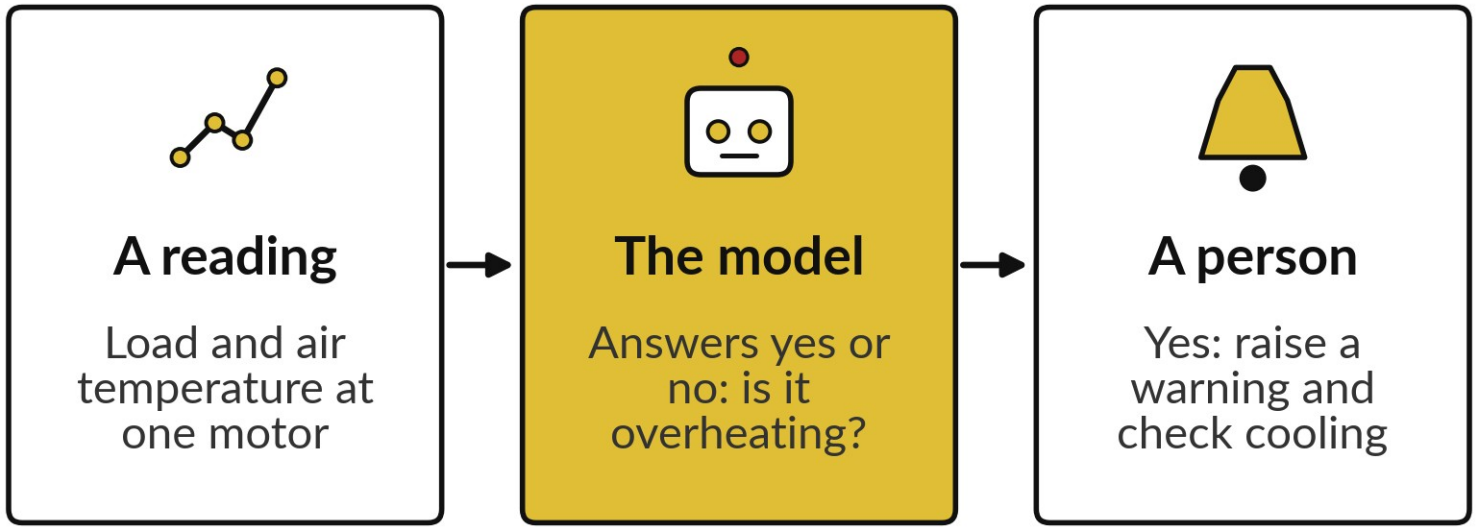
Question	Short answer
What is a baseline model for?	A baseline model is the simplest possible prediction, here the mean of the training target. Every model must beat it. If a model cannot, it has learned nothing useful.
What does the unit of the MAE tell you?	The mean absolute error (MAE) is in the unit of the target variable, so an MAE of 2 deg C means the prediction error is about 2 deg C on average.
Why do we split the rows by equipment group?	So that the validation set holds whole motor groups the model has never met, which is the situation in real use.

Check your own answer against these before we start today.

Words we use today

Word	What it means
Binary classification	Two possible answers: the positive class (yes, such as "overheating") and the negative class (no). The label is 1 or 0.
Decision threshold	The probability at or above which the model predicts positive (yes).
Confusion matrix	A table of actual against predicted classes: TP, FN, FP and TN (true or false, positive or negative).
Precision and recall	Precision: of the predicted positives, the share that are right. Recall (sensitivity): of the actual positives, the share found.
ROC curve and AUC	The ROC curve plots true positive rate against false positive rate. AUC, the area under it, is 0 to 1: how well positives rank above negatives.

Binary classification gives a yes or no answer that tells a maintenance team whether to act



Binary classification has two classes: yes (positive) and no (negative). The question: is it overheating?

In Class 1 (regression) the model gave a number, such as the motor winding temperature in deg C.

Today the model reads the features and gives one of two classes: yes or no.

Regression predicts a number and classification predicts a class

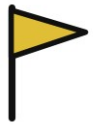


Regression

The answer is a number in deg C

Class 1: linear regression and a tree

Scored by prediction error: MAE and RMSE



Classification

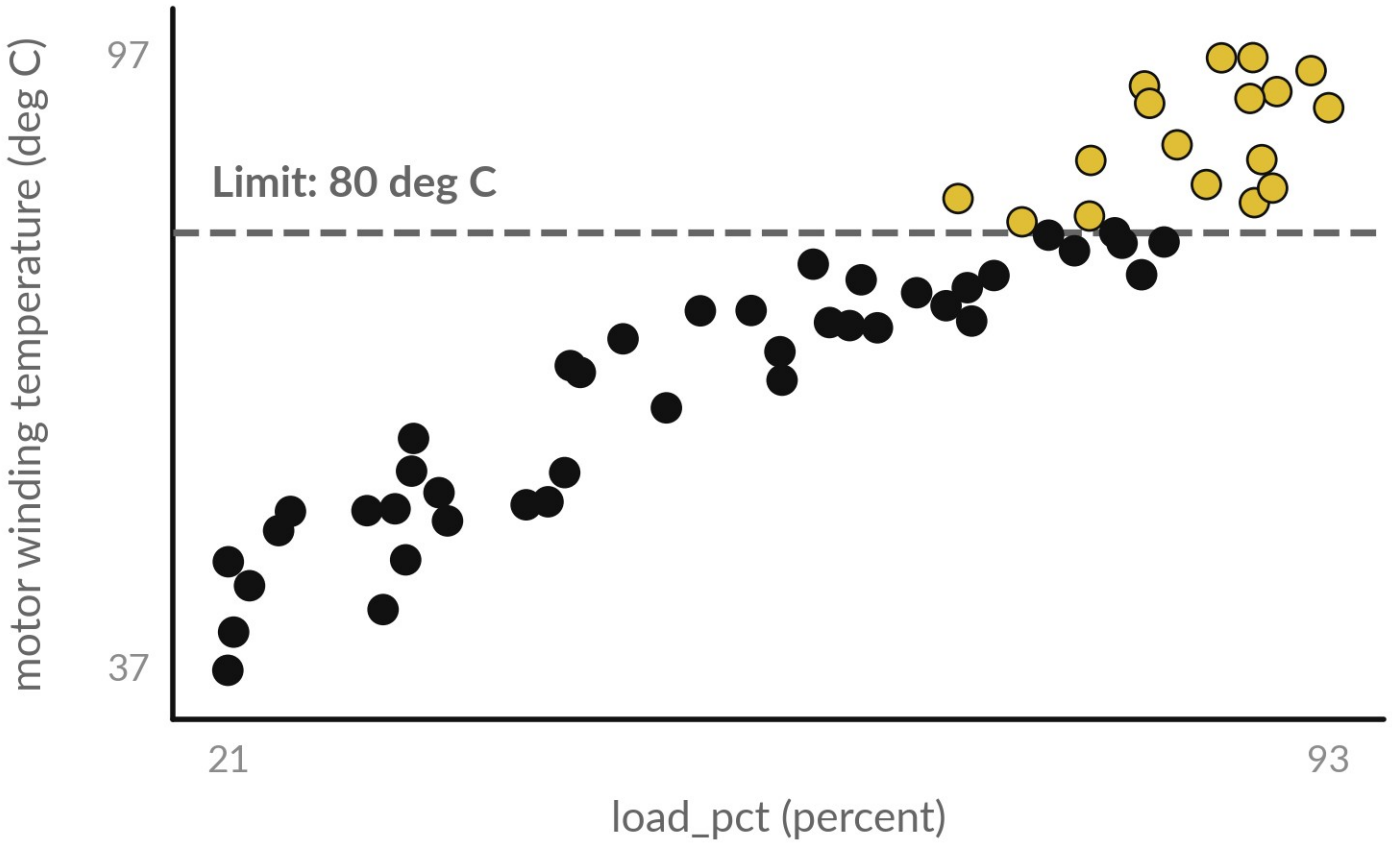
The answer is a class: positive or negative

Today: logistic regression and a decision tree classifier

Scored by counting TP, FN, FP and TN

Same data and same split as Class 1. Only the question changes.

A limit on the motor winding temperature turns each row into a positive or negative label



The label (target variable) is positive (1, yes) when the motor winding temperature is 80 deg C or more, and negative (0, no) below.

Each dot is one row of the training set. Gold dots are positive: 17 of 58.

Validation set: 3 of 15 rows are positive.

The features are still load_pct and ambient_c.

The majority-class baseline always answers no, is right on 80 percent of rows and finds no actual positive

Actual classes: 12 no, 3 yes



Majority-class baseline: always says "no"



Right on 12 of 15 rows. Accuracy 0.80.

Red: the 3 actual positives this baseline never finds.

A high accuracy that finds no positive is useless.

The majority-class baseline always predicts the more common class, here no. Accuracy is the share of rows answered correctly.

Always no is right on 12 of 15 validation rows: accuracy 0.80.

It finds none of the 3 actual positive rows.

A model must beat accuracy 0.80 and also find positive rows.

Standardisation (StandardScaler) puts load_pct and ambient_c on the same scale before the model learns

	A	B	C	D
	load_pct	ambient_c	load z-score	ambient z-score
1	59.15	31.71	-0.03	1.18
2	77.46	29.97	0.77	0.80
3	69.60	19.87	0.43	-1.42

Standardisation rewrites each feature as a z-score.

A z-score counts how many standard deviations (the typical distance from the mean) a value lies above or below the training mean.

Each table row is one training row (training row 1 (record_id 1) is first).

The z-score formula: subtract the training mean, then divide by the training standard deviation

Standardisation (z-score) $$x' = \frac{x - \mu}{\sigma}$$

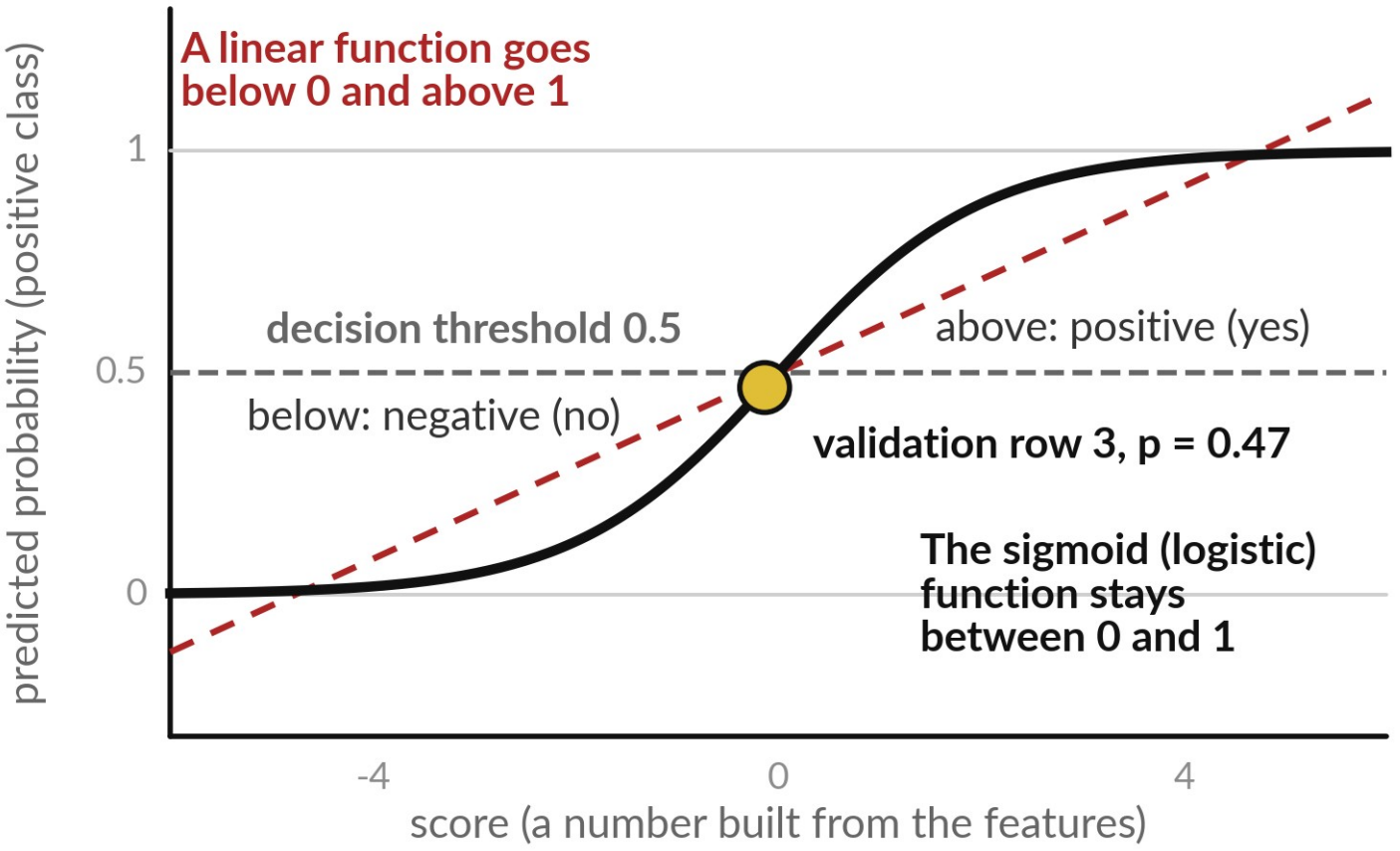
Training row 1, load_pct $$x'_1 = (59.15 - 59.89)/22.73 = -0.03$$

Training row 1, ambient_c $$x'_2 = (31.71 - 26.33)/4.55 = 1.18$$

Symbol	What it means	In our data
x	A raw feature value	Training row 1: 59.15 % and 31.71 deg C
x'	The standardised value (z-score): 0 is the mean, 1 is one standard deviation above	Training row 1: x1' = -0.03, x2' = 1.18
mu	The training mean of that column	load 59.89 %; ambient 26.33 deg C
sigma	The training standard deviation of that column	load 22.73 %; ambient 4.55 deg C

Subscript 1 is load_pct and 2 is ambient_c. The validation set is standardised with the same mu and sigma.

Logistic regression passes a linear score through the sigmoid (logistic) function, which stays between 0 and 1



The sigmoid (logistic) function turns any score into a probability from 0 to 1.

A probability of 0.5 means the two classes are equally likely.

The decision threshold is the probability at or above which the model predicts positive (yes), and below which it predicts negative (no).

Logistic regression in three lines: score, probability, prediction

1 Score

$$z = b_0 + b_1 x'_1 + b_2 x'_2$$

2 Probability

$$p = \frac{1}{1 + e^{-z}}$$

3 Prediction

$$\hat{y} = \text{positive if } p \geq t, \text{ otherwise negative}$$

Symbol	What it means	In our data
z	The score: any number	Validation row 3 (record_id 17): z = -0.13
b0	Intercept (bias): score at the mean	-1.99
b1, b2	Weights (coefficients): rise in z per standard deviation	b1 = 2.81 (load), b2 = 0.93 (ambient)
x1', x2'	Standardised load_pct and ambient_c	Validation row 3: 0.08 and 1.75
p	Probability of positive class. e = 2.718	Validation row 3: p = 0.47
t	Decision threshold: predict positive if p is at least t	0.5 first, then 0.3 and 0.7

Worked example: validation row 3 gets p = 0.47, so the model predicts no

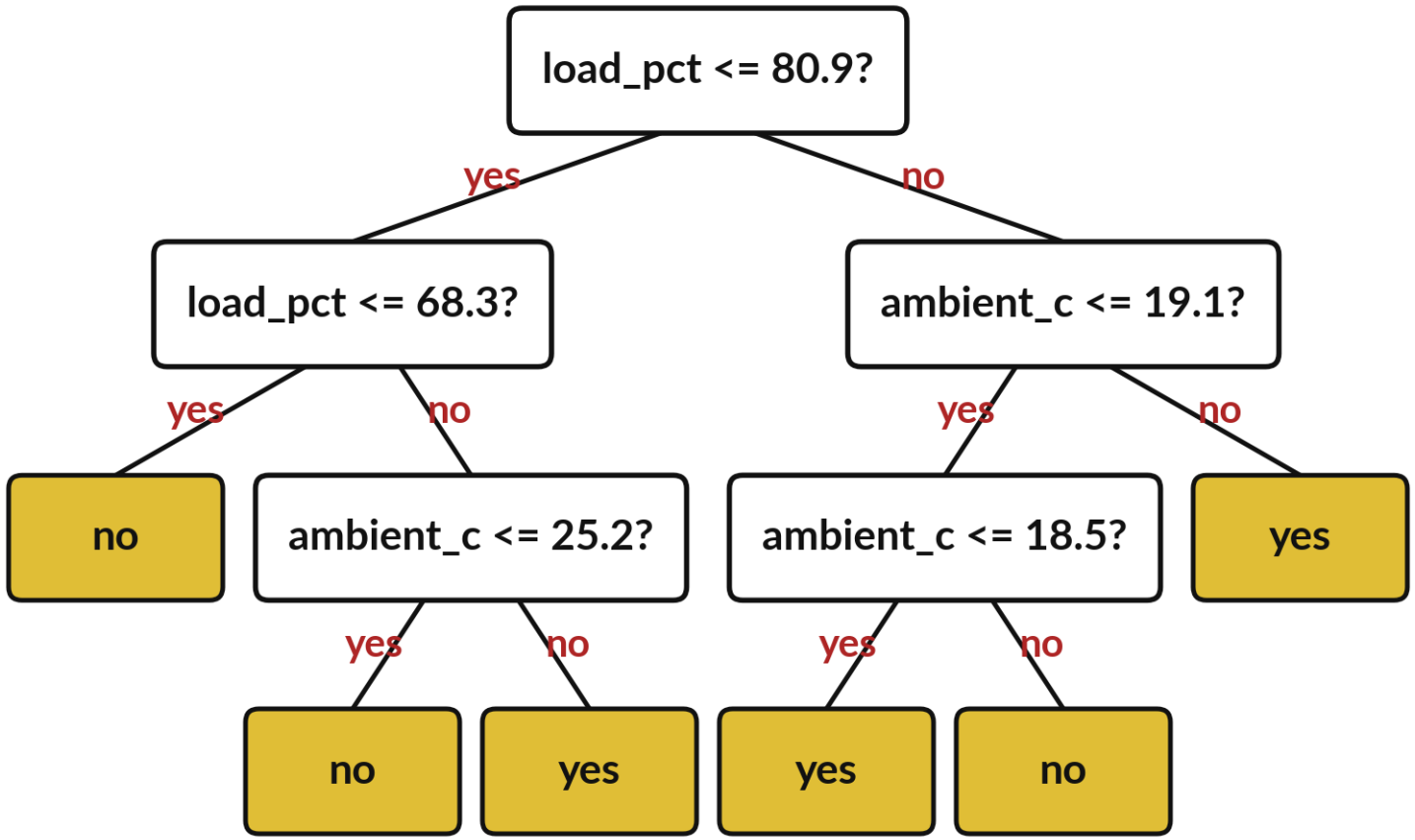
	A	B	C	D
	Part	**z-score x'**	**Weight b**	**Product**
1	Intercept b0			-1.99
2	load_pct	0.08	2.81	0.23
3	ambient_c	1.75	0.93	1.63
4	Score z (sum)			-0.13
5	Probability p			0.47

Validation row 3 (record_id 17): load_pct 61.76 %, ambient_c 34.28 deg C. Actual class: positive.

p = 0.47 is below the threshold 0.5, so the prediction is no: a false negative (FN).

One standard deviation more load_pct adds 2.81 to z. One more of ambient_c adds 0.93.

A decision tree classifier can also answer yes or no, and it does not need standardisation



A row follows the answers down the tree until it reaches one leaf node, a yellow box.

Each question is an internal node: a split on a feature.

A leaf node predicts yes when most training-set rows that ended there are yes.

Validation accuracy: 0.93.

How to read the tree

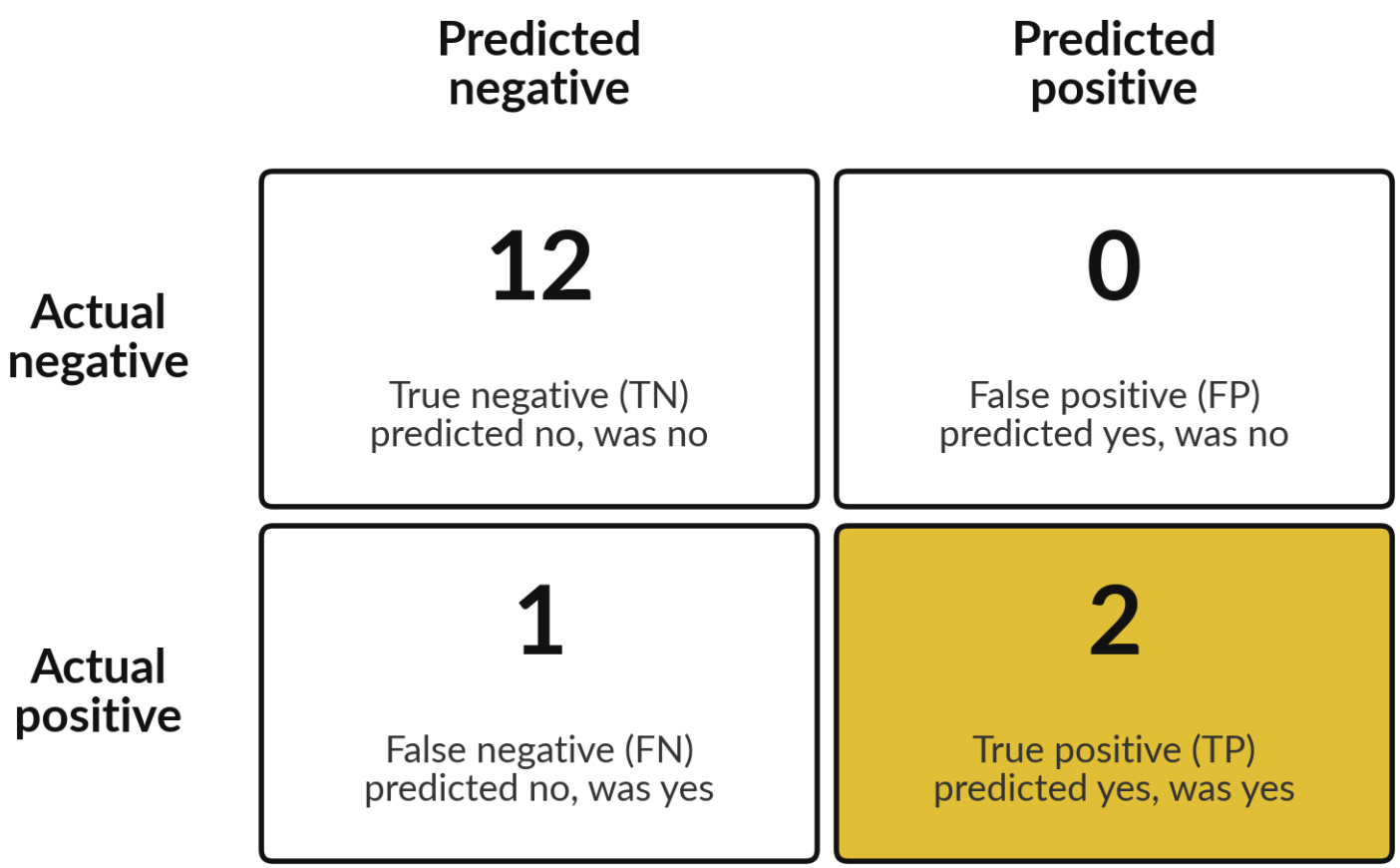
A leaf node (a yellow box) predicts the majority class of the training-set rows that ended there.

max_depth is a hyperparameter, a setting chosen before training. It is the largest number of questions (internal nodes) a row can be asked on the way down. Here it is 3.

A deeper tree can fit the training set so closely that it overfits: it fits the noise. Class 3 tests that.

The questions use the real feature values, so a tree needs no standardisation.

The confusion matrix counts true and false positives and negatives



Logistic regression, decision threshold 0.5, validation set. Positive class: overheating.

Each of the 15 rows lands in exactly one box.

Rows: Actual class. Columns: Predicted class, as in scikit-learn.

FP is a false warning. FN is a missed positive.

Accuracy and precision are worked out from the four counts TP, FP, FN and TN

Accuracy $$\text{accuracy} = \frac{TP + TN}{n} = \frac{2 + 12}{15} = 0.93$$

Precision $$\text{precision} = \frac{TP}{TP + FP} = \frac{2}{2 + 0} = 1.00$$

Symbol	What it means	In our data
TP	True positive: predicted positive, actually positive	2 validation rows
FP	False positive: predicted positive, actually negative	0 rows
FN	False negative: predicted negative, actually positive	1 row
TN	True negative: predicted negative, actually negative	12 rows
n	All rows scored: TP + TN + FP + FN	15 validation rows

Counts are for logistic regression at decision threshold 0.5.

Recall (sensitivity) and F1: how many actual positives we find, and one blended score

Recall

$$R = \text{recall} = \frac{TP}{TP + FN} = \frac{2}{2 + 1} = 0.67$$

F1

$$F_1 = \frac{2PR}{P + R} = \frac{2 \times 1.00 \times 0.67}{1.00 + 0.67} = 0.80$$

Symbol	What it means	In our data
R	Recall (sensitivity, true positive rate): share of actual positives found	2 found of 3 actual positive rows
P	Precision: the share of the model's positive predictions that are right	1.00 (previous slide)
F1	The F1 score, a blend of P and R. It is low when either one is low	0.80
TP, FN	True positives and false negatives (previous slide)	2 and 1

Same F1 from the counts: 2TP / (2TP + FP + FN) = 4 / (4 + 0 + 1) = 0.80.

The four scores in words: the question each one answers

	Score	The question it answers	By hand on these counts	Result
	**Accuracy**	Out of all rows, how often is the prediction right?	(2 + 12) / 15	0.93
	**Precision**	Of the rows the model predicted positive, how many are really positive?	2 / (2 + 0)	1.00
	**Recall (sensitivity)**	Of the rows that are really positive, how many did the model find?	2 / (2 + 1)	0.67
	**F1**	One number that blends precision and recall	2 x 1.00 x 0.67 / (1.00 + 0.67)	0.80

Counts: TP 2, FN 1, FP 0, TN 12. The majority-class baseline has accuracy 0.80.

A false negative and a false positive cost different things for a motor

False negative (FN)

Predicted no, but actually yes.

A winding overheats with no warning.

The insulation can be damaged.

Every FN lowers recall.

False positive (FP)

Predicted yes, but actually no.

Someone checks a healthy motor.

It costs some time.

Every FP lowers precision.

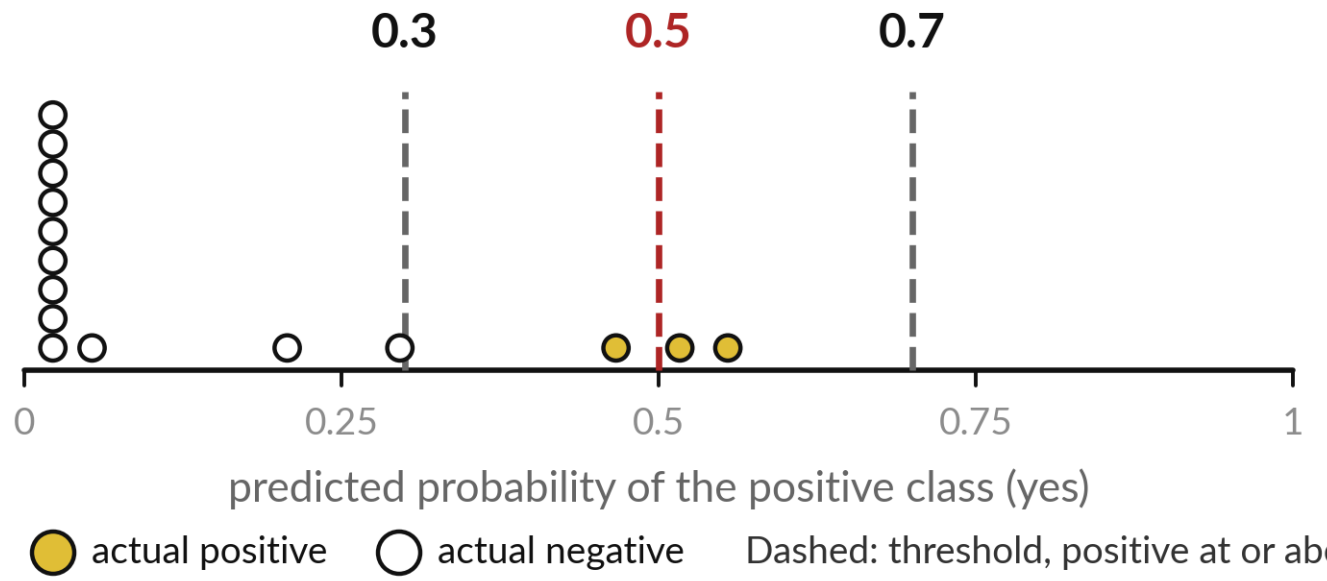
Here a false negative costs more, so we watch recall.

Both models give the same counts on the same 15 rows: TP 2, FN 1, FP 0

	Measure	Logistic regression	Decision tree, depth 3	Majority-class baseline
✅	True positives (TP)	2	2	0
❌	False negatives (FN)	1	1	3
🔔	False positives (FP)	0	0	0
📊	Accuracy	0.93	0.93	0.80
🚩	Precision	1.00	1.00	0.00
🔍	Recall	0.67	0.67	0.00
📈	F1	0.80	0.80	0.00

On these 15 rows the two models reach the same F1, so we cannot name a winner. Precision of 0.00 for the majority-class baseline means 0 of 0: it never predicts positive.

Moving the decision threshold changes recall, and on these 15 rows no threshold adds a false positive



Threshold 0.3: TP 3, FN 0, FP 0, recall 1.00

Threshold 0.5: TP 2, FN 1, FP 0, recall 0.67

Threshold 0.7: TP 0, FN 3, FP 0, recall 0.00

A lower decision threshold raises recall. Here it adds no false positive, but on other data it usually would.

Each dot is one validation row. Gold: actual positive. White: actual negative.

Validation row 3 (record_id 17) is an actual positive with p = 0.47: predicted positive at 0.3, negative at 0.5 and 0.7.

Each decision threshold gives its own counts and its own scores

	Threshold	TP	FN	FP	Precision	Recall	F1	Accuracy
🔔	**0.3**	3	0	0	1.00	1.00	1.00	1.00
✅	**0.5**	2	1	0	1.00	0.67	0.80	0.93
🚩	**0.7**	0	3	0	0.00	0.00	0.00	0.80

TP: true positives. FN: false negatives. FP: false positives. At 0.7 no row is predicted positive, so precision shows 0.00. 0.5 to 0.3: recall goes from 0.67 to 1.00; FP stays at 0.

On a larger set, a lower decision threshold finds more positives and also gives more false positives

	Decision threshold	TP (of 10 actual positive)	FP (of 90 actual negative)	True positive rate (TPR)	False positive rate (FPR)
🔔	**0.2**	10	27	1.00	0.30
✅	**0.5**	8	6	0.80	0.07
🚩	**0.8**	4	1	0.40	0.01

Illustrative example (hypothetical data), not from our data. TPR is the same as recall. Each row of this table becomes one point on a ROC curve.

Two rates give the points of the ROC curve: the false positive rate (FPR) and the true positive rate (TPR)

False positive rate (at 0.5)

$$\text{FPR} = \frac{FP}{FP + TN} = \frac{0}{0 + 12} = 0.00$$

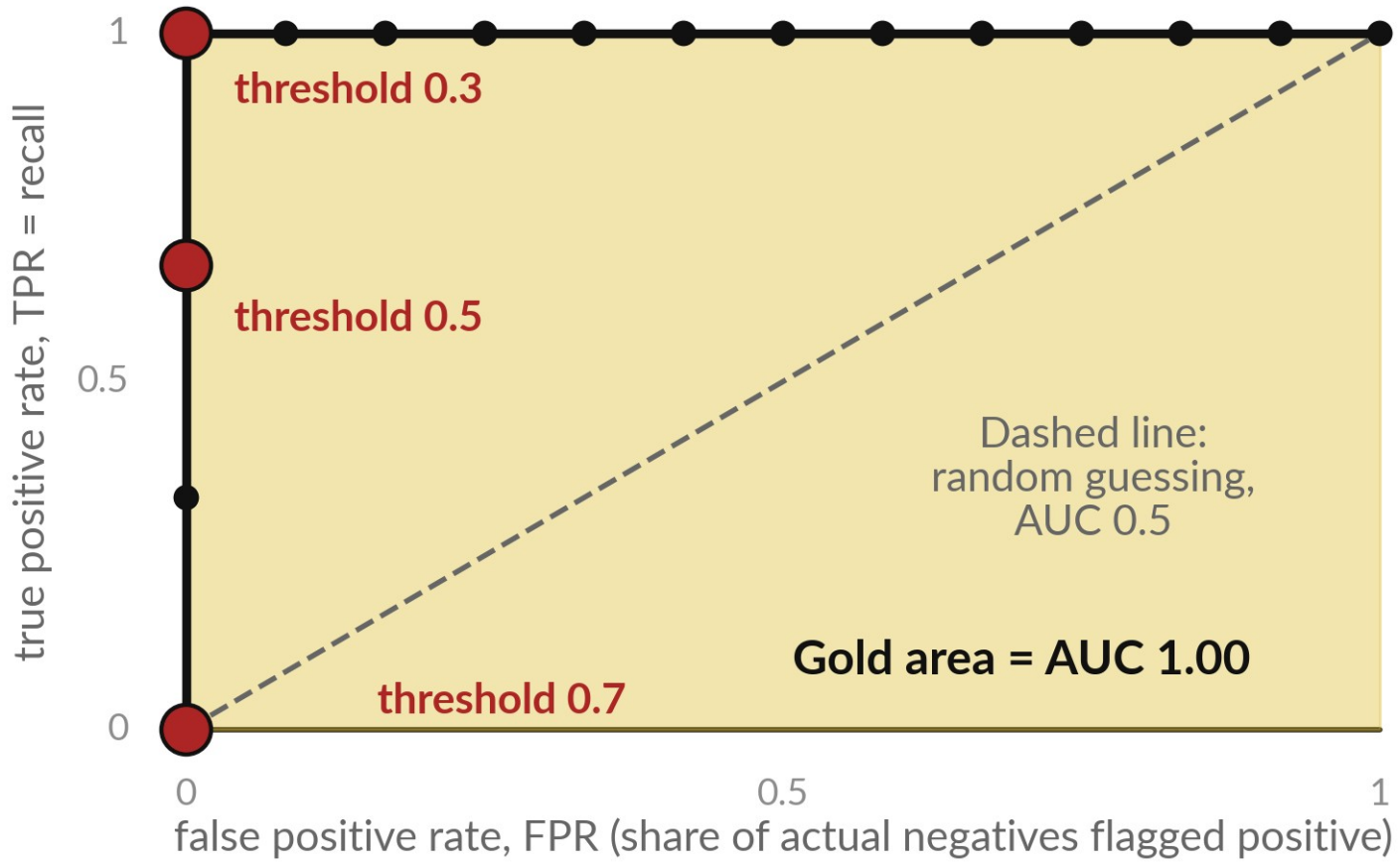
True positive rate (at 0.5)

$$\text{TPR} = \frac{TP}{TP + FN} = \frac{2}{2 + 1} = 0.67$$

Symbol	What it means	In our data
FPR	False positive rate: share of actual negatives (FP + TN) predicted positive	0 of 12 actual negative rows
TPR	True positive rate: share of actual positives (TP + FN) found. Same as recall	2 of 3 actual positive rows
Specificity	True negative rate: TN / (TN + FP). Equals 1 - FPR	12 of 12 = 1.00

Points (FPR, TPR) at thresholds 0.3, 0.5 and 0.7: (0.00, 1.00), (0.00, 0.67), (0.00, 0.00).

The ROC curve plots one point for every decision threshold: FPR across, TPR up

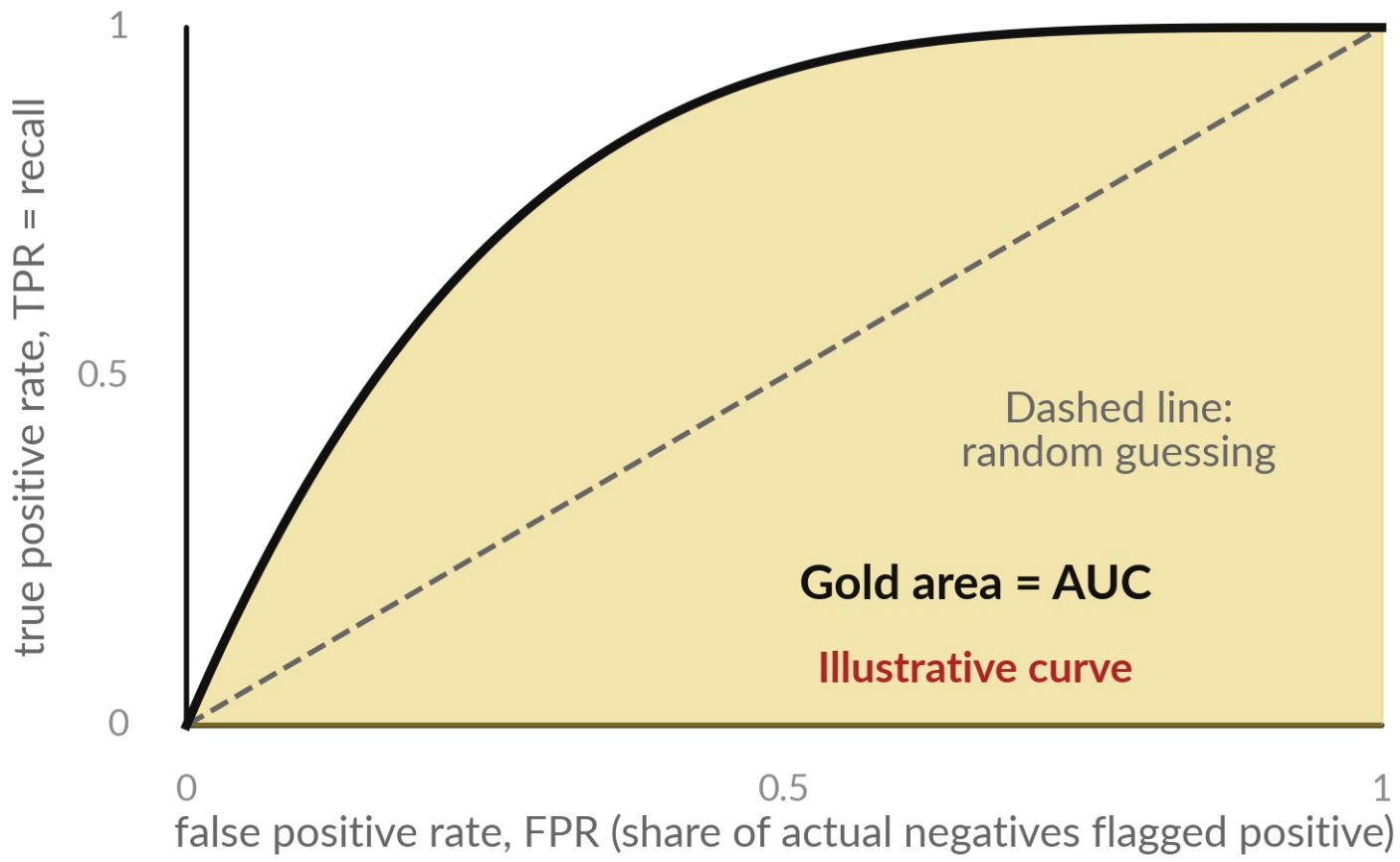


ROC is short for receiver operating characteristic. It is a graph with one point per decision threshold.

Each black dot is one threshold. 15 validation rows give only 16 points, so the line is a staircase.

The three red dots are our thresholds 0.3, 0.5 and 0.7 from the table. The gold area under the line is the AUC.

AUC is the area under the ROC curve, and on a large dataset the curve is smooth



AUC (area under the curve) is the chance that an actual positive row scores higher than an actual negative row.

Ours: 36 pairs (3 x 12), and 36 of 36 pairs are in the right order: AUC 1.00.

The curve is illustrative. With thousands of rows it looks smooth. AUC 0.5 is random guessing.

With 15 rows and 3 actual positives, every score here is shaky



The validation set has 15 rows and only 3 actual positive rows.



If one prediction changed, recall would move by 0.33 and accuracy by 0.07.



These numbers teach you how the scores work. They are too few to prove which model is better.



Class 3 shows how to test a model with more care.

Summary: judge a binary classifier by counting each kind of mistake



Make a label from the target with a limit: positive class or negative class.



Standardise the features. Logistic regression turns a score into a probability with the sigmoid function.



A decision threshold turns the probability into a predicted class.



Count TP, FP, FN and TN, then work out accuracy, precision, recall and F1.



Move the decision threshold, or read the ROC curve and AUC, to balance recall against precision.

Read the workflow before you write it: impute, standardise, fit, predict probabilities, count

```
imputer = SimpleImputer(strategy="median")
X_train = imputer.fit_transform(train_features)
X_valid = imputer.transform(valid_features)
model = make_pipeline(StandardScaler(),
                      LogisticRegression())
model.fit(X_train, y_train)
y_proba = model.predict_proba(X_valid)[:, 1]
threshold = 0.5
y_pred = (y_proba >= threshold).astype(int)
cm = confusion_matrix(y_valid, y_pred)
```

In plain words

The imputer fills each missing value (NaN) with the training median (imputation).

The pipeline chains StandardScaler and LogisticRegression.

predict_proba gives a probability per class. Column 1 is the positive class.

y_proba >= threshold gives True or False. astype(int) makes it 1 or 0: y_pred.

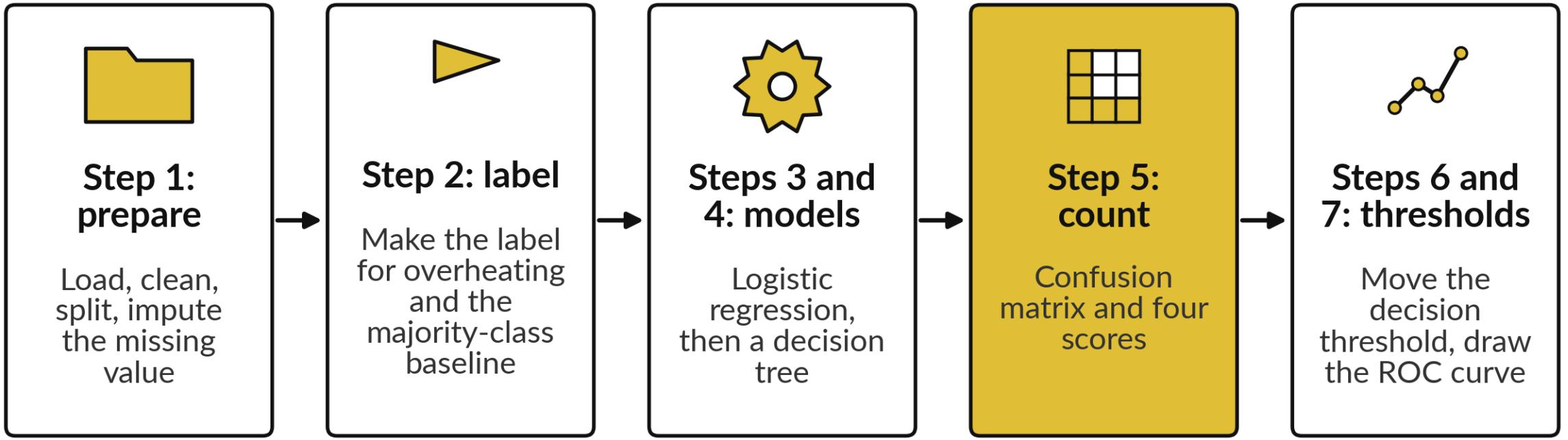
confusion_matrix puts Actual down the side and Predicted across the top, negative first: TN and FP in the top row, FN and TP below.



Part 2: the lab

Close the slides. Open the notebook. You use the same engineering_data.csv file as in Class 1.

The lab notebook has five stages, and each stage ends with a check



Predict first, run second, then read the output. CHANGE tasks: steps 3 and 6. Steps 8 and 9: conclusion and saving.

Check your numbers against these before you move on

	What to check (validation set)	You should see
	Positive labels: training set, validation set	17 of 58, 3 of 15
	Majority-class baseline accuracy	0.80
	Learned weights (coefficients): load_pct, ambient_c, and the intercept	2.81, 0.93, -1.99
	Logistic regression: TP, FN, FP, TN	2, 1, 0, 12
	Logistic regression: accuracy, precision, recall, F1	0.93, 1.00, 0.67, 0.80
	Decision tree: TP, FN, FP, TN	2, 1, 0, 12
	Logistic regression recall at decision thresholds 0.3, 0.5, 0.7	1.00, 0.67, 0.00
	AUC	1.00

If your number differs, stop and ask. Do not change a number to match. Find the step that went wrong.

Four messages you will meet, what they mean and what to do

	Message	What it says	What to do
	**ValueError: contains NaN**	A missing value (NaN) reached StandardScaler or the model	Run the imputation step (SimpleImputer with the training median)
	**Classification metrics can't handle a mix of binary and continuous targets**	You gave probabilities where a predicted class (0 or 1) was needed	Compare y_proba with the threshold first, to get y_pred
	**ConvergenceWarning**	The model stopped before it finished learning. The cell still runs	Standardise the features, or raise max_iter
	**NameError**	A cell above has not been run yet	Run all cells from the top, in order

How this class counts towards your coursework



Block 1 is Class 1 and Class 2.



Assignment part A (A7, regression) comes from the Class 1 lab. Part B (A8, classification) comes from today's lab.



The Block 1 assignment closes on Friday 20 November, 11:00 (Kampala time).



A short quiz of five questions covers both classes. It opens in the last 20 minutes of today's class and stays open for 24 hours.

Save the notebook and keep your scores

What you keep from today

Your lab notebook, saved with all cells run in order, and the scores from the confusion matrix and the thresholds.

Our question today

Can a binary classifier say yes or no to "overheating", and how do we count its true and false positives and negatives?

Next class

Class 3: can I trust this model? We test it more carefully.

Take away

Saved notebook
Restart and run all, then save

Confusion matrix
TP, FN, FP and TN

Scores
Four scores and AUC

Bring next time
The same laptop and notebook